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ASSIGNMENT · 3

Name - Anjali Tomar

Roll No. - 2401010080

Section - A

Subject - Database management system.

Ques ① What is SQL? Why is it important in DBMS?
SQL (structured query language) is a standard programming language that is used to create, manage, store and manipulate the data in databases.
It is important because-
(i) it allows efficient data handling
(ii) it allows quick retrieval of data using queries.

Ques ② What is the difference b/w DDL and DML?

DDL	DML
• (Data definition language).	• (Data manipulation lang.)
• It is used to define and modify the structure of database	• It is used to manipulate the data stored in the database tables.

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Ques-③ What is a join? Explain its types.

A join in SQL is used to combine data from two or more tables based on a related column between them. It helps retrieve meaningful info. by connecting tables.

Types -

(i) Inner join

(ii) Left join

(iii) Right join

(iv) Full join

Ques-④ What is "GROUP BY" and "HAVING" clause?

These are used in SQL to work with grouped data.

- GROUP BY - used to group rows.
- HAVING - used to filter the grouped data

Ques-⑤ What is a subquery?

A subquery is a query that is nested inside another SQL query. It is used to perform complex queries/operations by first executing the inner query and then using its result in the outer query.

Ques-⑥ What is relational algebra?

Relational algebra is a procedural query language used in DBMS to retrieve and manipulate data from relational databases.

Some common operations includes-

- Selection (σ)
- projection (π)
- Join ($\bowtie$)
- Set difference (−)

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Ques (7) What is the difference b/w WHERE & HAVING.

WHERE	HAVING
• It is used to filter rows before any grouping is done.	• It is used to filter data after the GROUP BY operation.

Section - B

Ques (8) Write an SQL query using to retrieve all standard students with marks >75.

```
SELECT *
FROM students
WHERE marks >75;
```

Ques (9) Write a query using JOIN to combine students and course Tables.

```
SELECT student.student_name, course.course_name
FROM students
INNER JOIN course
ON student.course_id = course.course_id;
```

Ques (10) Write a query using GROUP BY to find students in a specific department.

```
SELECT department, COUNT(*) AS
total_students
FROM students
GROUP BY department;
```

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Section - C

Ques (11) Write a query to find employee in a specific dept.

```
SELECT e.employee_name, d.department_name
FROM employee
INNER JOIN department d
ON e.department_id = d.department_id
WHERE d.department_name = 'sales';
```

Ques (12) An e-commerce system stores orders. Retrieve total sales per customer.

```
SELECT customer_id, SUM (order_amount) AS total_sales
FROM orders
GROUP BY customer_id;
```

Ques (13) A university database needs top 3 students. Write a query.

```
SELECT *
FROM students
ORDER BY marks DESC
```

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Section-D.

Ques (14) Write a SQL query for any real life system and explain logic.

```
SELECT name, salary
FROM employees
WHERE salary > 50000;
```

Ques (15) Write a JOIN query and mention one use case

```
SELECT o.order_id, c.customer_name
FROM orders o
INNER JOIN customers c
ON o.customer_id = c.customer_id;
```

Use-case - used in e-commerce system to display combine customer placed each order.

Ques (16) Write a GROUP BY query and explain its output.

```
SELECT department, COUNT(*) AS total_employees
FROM employees
GROUP BY department;
```

explanation - This query groups employees by department and counts how many employees are in each department. The output will show each department along with the total number of employees in it.

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